Status	Finished
Started	Wednesday, 28 October 2026, 9:34 PM
Completed	Wednesday, 28 October 2026, 9:39 PM
Duration	4 mins 35 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Inserting data into a list
- b. Arranging data in a particular format
- c. Searching for data in a list
- d. Deleting data from a list

The correct answer is: Arranging data in a particular format

Question **2**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It sorts data using a single pass
- c. It does not require recursion
- d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **3**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Insertion
- b. Distribution
- c. Extraction
- d. Selection

The correct answer is:
Insertion

Question **4**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Linear Search
- d. Binary Search

The correct answer is: Merge Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are discarded
- c. They are split again into smaller sublists
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. sort()
- b. arrange()
- c. sorted()
- d. order()

The correct answer is: sort()

Question **9**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Dynamic programming
- c. Greedy method
- d. Backtracking

The correct answer is:
Divide and conquer

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting is rarely used in programming
- b. Sorting decreases the efficiency of selection operations
- c. Sorting makes it harder to search for items
- d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **11**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Insertion Sort

The correct answer is: Bubble Sort

Question **12**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It simplifies the sorting process by using only one pass
- c. It only works on small datasets
- d. It avoids the need for recursion

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **13**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort does not guarantee sorted order
- b. Bubble Sort cannot handle duplicate elements
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Selection sort
- c. Quick sort
- d. Merge sort

The correct answer is:
Merge sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. sort()
- b. arrange()
- c. sorted()
- d. order()

The correct answer is: sorted()